Digital Audio-Video Signal Processing

Prof. Marco Tagliasacchi

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1 Question - 7 pts

- Define the entropy rate of a discrete source.
- State the Shannon's coding theorem for a discrete source with memory.
- Illustrate one practical coding technique that can be used to approach the Shannon's bound. How is the source memory taken into account? What can be done when the source statistics are time varying?

2 Question - 7 pts

- Illustrate the block diagram of the DPCM coding scheme.
- Explicitly show that the quantization error in the *prediction error signal* domain is the same as in the *original signal* domain.
- Explain a few strategies that can be adopted when the statistics of the input signal are time varying.

3 Question - 6 pts

Illustrate the basic principles behind the MDCT. Illustrate how the MDCT is able to accommodate signals with non-stationary time-frequency characteristics.

4 Question - 12 pts

Let $\mathcal{U}(x; a, b)$, $-\infty < a < b < +\infty$ denote a uniform pdf defined by

$$\mathcal{U}(x; a, b) = 1/(b-a) \text{if } a \leq x < b, \text{ 0 otherwise} \tag{1}$$

Consider a source that emits symbols from an infinite alphabet according to the following rules

- for each i, $c(i) \sim \mathcal{N}(0, 1)$
  - if $c(i) \geq 0$, $x(2 * i) \sim \mathcal{U}(0, 1)$ and $x(2 * i + 1) \sim \mathcal{U}(0, 1)$
  - else, $x(2 * i) \sim \mathcal{U}(-1, 0)$ and $x(2 * i + 1) \sim \mathcal{U}(-1, 0)$

Perform the training of the codebook of a 0.5-bit/symbol vector quantizer that encodes each pair of consecutive symbols jointly.
Compute the quantization MSE.